

OmegaSelf-Compatible Emotion: From Psi Modulation to Auditable Cognitive Regimes

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19 July 2026

Abstract

We develop a functional theory of emotion for self-modeling cognitive agents and multi-agent hives. The motivating question is not which human emotion labels an artificial agent should imitate, but which persistent, coordinated changes of cognition improve its goals given its actual actuators. The proposed architecture has three levels. First, a continuous Psi-style motivational substrate computes need dynamics, valence, arousal, competence, resolution, selection, and securing. Second, evidence-grounded appraisals are evaluated relative to *OmegaSelf*, an explicit model of goals, capabilities, commitments, provenance, resources, and predicted behavior. Third, optional named regime latches provide temporal abstraction, hysteresis, inspectable identity, bounded actuator bundles, and governance hooks. A regulator converts all emotional effects into clamped configuration requests; it does not grant authority.

We formalize appraisals, behavioral actuator support, continuous modulation, regime activation, composition, value, and auditability. We prove qualified results concerning actuator nullity, return to baseline, contraction versus self-exciting lock-in, switching-cost hysteresis, convex safety of blended regimes, and information-theoretic legibility. These results correct several tempting but invalid stronger claims: palette size is not generally bounded by the number of actuators; positive switching cost does not imply every optimal policy has the same hysteresis geometry; and continuous modulation alone may already implement causally potent, object-directed emotion. The discrete regime layer must therefore earn its extra ontology empirically.

We relate the framework to appraisal, action-readiness, basic-emotion, constructionist, core-affect, predictive-processing, cybernetic, social-functional, and emotion-regulation theories. We inspect MetaMo/OpenPsi at pinned commit 102da717b5354d6fd49d9226d92a32091438bc94: its continuous appraisal and fuzzy feeling layers provide a working substrate but not the proposed persistent hysteretic latch. Finally, we specify research-assistant and cybersecurity palettes, an OmegaClaw implementation seam, safety invariants, and controlled experiments comparing continuous Psi alone against continuous Psi plus regimes. The account concerns functional emotion; it neither assumes nor establishes phenomenal feeling.

1 Status, scope, and provenance

This paper systematizes a July 2026 discussion about emotion, *OmegaSelf*, Hyperseed, ECAN, MetaMo/OpenPsi, and OmegaClaw. It is a theory and engineering proposal, not a report of a deployed emotional controller. Source inspection supports claims about the current MetaMo and OmegaSelf artifacts; behavioral claims about regime latches remain hypotheses until tested.

We use the following epistemic convention:

- **Observed:** directly inspected in a named source or reproduced test.

- **Derived:** follows mathematically from explicit assumptions.
- **Hypothesis:** plausible and falsifiable, but not established.
- **Design decision:** a recommended implementation constraint.

The Bach tutorial used here is a 132-slide architectural tutorial [1]. Its equations are a valuable reference model, but the tutorial alone does not establish that each equation is empirically optimal. Claims of MicroPsi fidelity require a separately pinned implementation and conformance tests. The MetaMo source examined for this paper was pinned and its sixteen upstream MeTTa test files, including OpenPsi appraisal and feeling tests, passed locally through PeTTa on 19 July 2026.

2 The design problem: emotion relative to an ecology

Animal emotions were selected in an ecology containing bodies, predators, mates, coalitions, pain, and rapid action. Anger is useful when a coordinated shift in posture, attention, pain tolerance, risk appetite, signaling, and force can remove an obstacle. Copying the label into a research assistant does not copy the ecology or the action repertoire.

The appropriate abstraction is:

An emotion is a temporally extended, self-relevant control organization that coordinates multiple cognitive or bodily processes in preparation for a class of situations.

This yields an actuator-relative design question. A research hive can allocate attention, change query breadth, spawn critics, invoke proof tools, adjust verification, defer publication, consolidate memory, revise a representation, or ask a human. A cybersecurity hive can change monitoring, provenance demands, permissions, containment, red-team search, escalation, and report fusion. A military robot may possess rapid physical and tactical actuators for which an anger-like attack regime is meaningful. The palettes should differ because the actuator ecologies differ.

This is not equivalent to saying that each actuator requires one emotion. Many emotions may share actuators but differ in object, appraisal, time scale, and coordination pattern. Conversely, a single continuous controller may coordinate many actuators without any discrete emotion object. The central empirical issue is whether a named temporal macrostate improves control, learning, or governance.

3 Explanatory layers and psychological context

Major emotion theories often address different projections of a common control problem rather than supplying mutually exclusive full architectures.

Tradition	Principal contribution	Mapping in this framework
Appraisal (Lazarus, OCC, Scherer)	Goal relevance, novelty, agency, coping, norms, and sequential checks	The evidence-bearing appraisal map relative to <i>OmegaSelf</i> ; especially its object, attribution, controllability, and expected goal consequences [9, 10, 14].
Action readiness (Frijda)	Emotions organize readiness to transform a relation with the world	The behavior-changing support of an actuator bundle [6].

Tradition	Principal contribution	Mapping in this framework
Basic emotions and affect programs	Conserved, coordinated response families suited to animal bodies	Substrate-specific regimes; useful analogies, not a mandatory palette for software agents [5, 12].
Core affect	Low-dimensional valence and arousal vary continuously	A projection of the continuous modulator state, not the whole regime [13].
Psychological construction	Named categories depend on concepts, context, and learned categorization	A label or coarse-graining map over appraisal, modulation, object, and context [2].
Somatic marker and predictive accounts	Internal state and predicted bodily or control consequences bias choice	OmegaClaw telemetry and anticipated need/goal change provide functional interoception and prospective value [4, 15].
Cybernetic self-regulation	Affect tracks discrepancy and rate of progress toward reference values	Need depletion, goal-progress error, competence, and controller gain [3].
Emotion regulation	Selection, attention, reappraisal, and response can be regulated at different stages	A higher-order regulator changes appraisal, activation, blending, or admissible actuator requests [7].
Social-functional accounts	Emotions coordinate dyads, groups, status, and norms	Trust, communication, report-fusion, escalation, and signaling coordinates in a hive [8].
Psi/MicroPsi	Needs and modulators jointly organize perception, cognition, action, valence, and object-directed emotion	The continuous dynamical substrate under the optional regime layer [1].

The framework separates four objects that are frequently conflated:

1. a continuous control state (for example, valence and arousal);
2. an object-directed appraisal (for example, an attributed removable blocker to a valued goal);
3. a temporally extended functional regime that changes behavior; and
4. a linguistic category or report such as “frustration.”

None of these by itself establishes phenomenal consciousness. The rigorous claim for OmegaClaw is functional: the system may instantiate a fear-analogous or doubt-analogous control organization without thereby being shown to feel fear or doubt.

4 What Psi contributes, and what it does not lack

Bach’s tutorial derives affect from need and modulation dynamics rather than installing one independent variable for every named emotion [1]. Relevant mechanisms include need gain, loss, decay, and sensitivity; pain from depletion; urge and urgency; marginal aggregation of pleasure and pain; competence; arousal; resolution; selection; securing; anticipated reward; certainty; skill; and temporal discounting.

The tutorial also gives emotion objects to cognition and valence to motivation. Fear, anxiety, anger, and sadness are not merely decorative words. They are object-directed configurations that alter attention and action readiness. In particular, the distinction between anger and sadness depends on attribution and reachability: an attributed obstacle with a conceived response supports an anger-like organization, whereas failure with no conceived path supports a different organization. Therefore the proposed regime layer should not be advertised as making a previously noncausal Psi label causal.

The sharper synthesis is:

- **Psi**: continuous need, value, competence, and modulation dynamics already capable of object-directed emotional control;
- **OmegaSelf**: explicit, evidence-grounded, revisable appraisals of goals, commitments, capabilities, provenance, and predicted behavior;
- **regime latch**: an optional temporal abstraction that adds a stable macrostate identity, hysteresis, audit hooks, governed actuator requests, and a unit for credit assignment.

Thus a regime latch is an engineering hypothesis. If continuous Psi produces equal or better utility, calibration, recovery, and auditability, named emotions should remain descriptive projections. If latches improve temporally coherent control under switching costs and bounded oversight, they have earned the extra ontology.

5 Why OmegaSelf changes the appraisal problem

OmegaSelf treats the self-model as an event-sourced, continuously tested, defeasible theory rather than privileged introspection or autobiography [11]. Its central loop is evidence to belief to prediction to proposal to policy to action/probe to receipt to discrepancy to repair. It tracks contextual capability, commitments, evidence closure, prediction, continuity, and governance. A self-report is not authority, and no symbolic rule directly executes a skill; it emits a typed proposal consumed by a separate policy gate.

This is unusually compatible with a functional emotion architecture. Appraisal is not a free-form feeling generated by the language model. It is a query over versioned evidence about the agent and situation. “Goal stall,” for example, should identify a goal, expected progress curve, observed discrepancy, relevant method, evidence quality, available alternatives, and current capability beliefs. The resulting emotion request then passes through the same governance seam as any other proposal.

An objectful appraisal record should minimally contain

$$r_t = (n, g, o, a, q, \mathcal{A}, y, h), \quad (1)$$

where n is the affected need, g the goal, o the object or blocker, a a causal attribution, q controllability or coping potential, $\mathcal{A}$ the available alternatives, y the anticipated outcome distribution, and h the evidence closure and provenance. This structure distinguishes:

- a removable blocker from open-ended failure;
- epistemic uncertainty from task failure;
- adversarial obstruction from ordinary tool unreliability;
- low capability from insufficient resources; and

- genuine novelty from missing or stale evidence.

The self-model is important but should not be made logically necessary by definition. A reactive controller can have useful affect-like dynamics without an explicit self-model. The stronger, testable claim is that evidence-grounded self-indexing improves calibration and transfer when goals, capabilities, or environments shift (Conjecture 1).

6 Formal architecture

6.1 Spaces and kernels

Let W be the world-state space, Ω the OmegaSelf state space, $N \subset \mathbb{R}^p$ a need state, $M \subset \mathbb{R}^d$ the continuous modulator space, $C \subset \mathbb{R}^k$ the cognitive configuration space, A the action space, and Y the outcome space. A configuration $c \in C$ includes actual controllable coordinates such as ECAN wages, diffusion and rent parameters, verification depth, search temperature, subagent spawning, communication rate, memory consolidation, and tool-request thresholds.

Let

$$K(\cdot \mid c, w, \omega) \quad (2)$$

be the induced distribution over observable behavior and internal decisions. This kernel matters more than syntactic parameter movement: changing a parameter that is disconnected, saturated, or overridden is behaviorally null.

The continuous substrate evolves as

$$n_{t+1} = F_N(n_t, w_t, \omega_t, a_t, \xi_t), \quad (3)$$

$$m_{t+1} = F_M(m_t, n_{t+1}, r_t, \xi_t), \quad (4)$$

where ξ_t represents noise or unmodeled inputs. The appraisal is

$$r_t = \alpha(\omega_t, w_t, n_t, m_t). \quad (5)$$

Let $\mathcal{E}$ be a finite set of candidate regime types. Each regime has an activation $z_{e,t} \in [0, 1]$ with stateful update

$$z_{e,t+1} = H_e(z_{e,t}, r_t, m_t; \theta_e). \quad (6)$$

The parameters θ_e include entry and exit criteria, decay, refractory periods, and coupling. Hysteresis means the exit boundary differs from the entry boundary; it need not mean permanent stickiness.

6.2 Regimes, support, and governed composition

Definition 1 (Functional regime). *A functional emotional regime is a tuple*

$$e = (R_e, H_e, \Delta_e, \lambda_e, \mathcal{L}_e), \quad (7)$$

where R_e specifies admissible objectful appraisals, H_e the activation dynamics, $\Delta_e : C \rightarrow \mathbb{R}^k$ a requested configuration displacement, λ_e a decay or recovery parameter, and $\mathcal{L}_e$ a legibility record containing trigger, object, evidence, activation, requested effects, and predicted consequences.

Let B map continuous modulation into configuration requests. Before policy enforcement, the aggregate request is

$$u_t = Bm_t + \sum_{e \in \mathcal{E}} z_{e,t} \Delta_e(c_0). \quad (8)$$

A governor G converts this into a live configuration:

$$c_t = G(c_0, u_t, \omega_t, P_t), \quad (9)$$

where P_t is the active policy. G clamps magnitude, resolves conflicts, preserves invariants, and rejects unauthorized changes. An emotional regime is therefore a source of proposals, never a source of authority.

Definition 2 (Behavioral actuator support). *For a regime e at context (c, w, ω) , define*

$$\text{supp}_K(e; c, w, \omega) = \{i : \exists z > 0, K(\cdot \mid G(c, z\Delta_e, \omega, P), w, \omega) \neq K(\cdot \mid c, w, \omega)\}. \quad (10)$$

The support is behavioral, not merely the set of coordinates named by Δ_e .

Two regimes may have identical support but different objects or displacement vectors. Thus $|\mathcal{E}| \leq k$, $|\mathcal{E}| \leq |A|$, and similar simple palette bounds are false without additional injectivity assumptions.

6.3 Value and selection

For context class κ , horizon T , utility U_T , switching count S_T , and audit/control cost D_T , define the interventional regime value

$$V(e, \kappa) = \mathbb{E}[U_T \mid do(e), \kappa] - \mathbb{E}[U_T \mid do(e=0), \kappa] - \eta \mathbb{E}[S_T] - \gamma \mathbb{E}[D_T]. \quad (11)$$

The *do* notation is important. Activated regimes occur in difficult contexts, so observational correlation between activation and poor outcome can be misleading. Estimation should use randomized shadow/advisory trials when safe, or explicit causal adjustment with logged propensity and context.

Selection can weaken, specialize, merge, or remove a regime whose posterior value is nonpositive. Removal is not logically forced when value is exactly zero; it follows only under a complexity prior, maintenance cost, or explicit pruning rule.

7 Qualified formal results

7.1 No lever, no instrumental value

Proposition 1 (Behavioral actuator nullity). *Fix a context class κ . If the governed request of regime e leaves the behavior kernel unchanged almost surely throughout the evaluation horizon, and if e does not alter observation or utility directly, then its causal utility difference is zero. If maintaining or switching the regime has strictly positive expected cost, then $V(e, \kappa) < 0$.*

Proof. Kernel equality gives the same distribution of action and outcome trajectories under $do(e)$ and $do(e=0)$. Hence the first two terms of Equation 11 cancel. The remaining switching and audit/maintenance cost is nonnegative, and it is strictly positive by assumption. $\square$

This is the defensible form of “no lever, no emotion.” It establishes instrumental redundancy, not metaphysical impossibility. A behaviorally null state might still be a report, correlate, or phenomenal state; it simply has no demonstrated control value in the specified context.

7.2 Return to baseline

Theorem 1 (Return to baseline without re-evocation). *Suppose $\mathcal{E}$ is finite, $m_t \rightarrow m_0$, every activation $z_{e,t} \rightarrow 0$ after a final trigger, all requested displacements are bounded, and G is continuous at $(c_0, 0, \omega_\infty, P_\infty)$ with $G(c_0, 0, \omega_\infty, P_\infty) = c_0$. Then $c_t \rightarrow c_0$.*

Proof. Finiteness and bounded displacements imply $\sum_e z_{e,t} \Delta_e(c_0) \rightarrow 0$. Also $B(m_t - m_0) \rightarrow 0$ after absorbing Bm_0 into the baseline. Continuity of G then yields $c_t \rightarrow c_0$. $\square$

Decay alone is insufficient if the environment continually re-evokes the regime, the modulator baseline has shifted, or the governor contains its own memory. The theorem makes these exclusions explicit.

7.3 Self-excitation and lock-in

Theorem 2 (Linear contraction bound). *Let a regime activation vector satisfy*

$$z_{t+1} = \text{clip}(Az_t + \eta_t), \quad (12)$$

where clipping is nonexpansive in a norm $\|\cdot\|$, $\|A\| = q < 1$, and $\|\eta_t\| \leq b$. Then

$$\|z_t - z'_t\| \leq q^t \|z_0 - z'_0\| \quad (13)$$

for trajectories with identical input, and

$$\limsup_{t \rightarrow \infty} \|z_t\| \leq \frac{b}{1-q} \quad (14)$$

up to the clipping bound.

Proof. Nonexpansiveness gives the first inequality by induction. Iterating $\|z_{t+1}\| \leq q\|z_t\| + b$ yields $\|z_t\| \leq q^t \|z_0\| + b(1 - q^t)/(1 - q)$. $\square$

For the scalar update $z_{t+1} = (1 - \lambda + g)z_t + \eta_t$, contraction follows when $|1 - \lambda + g| < 1$, not merely from the slogan $\lambda > g$ in every possible parameterization. If $1 - \lambda + g > 1$ and positive excitation persists, clipping can create a high-activation attractor. But spectral radius above one alone does not guarantee paranoia under arbitrary signed inputs, resets, or policy intervention. The engineering requirement is to estimate the closed-loop gain, monitor occupancy and recovery, and preserve a safety margin in the actual nonlinear system.

7.4 Why switching costs can create hysteresis

Proposition 2 (A minimal hysteresis band). *Consider two regimes, 0 and 1, with a scalar sufficient statistic x . Let $D(x)$ be the expected continuation advantage of regime 1 over regime 0 before switching costs, continuous and strictly increasing, and let each switch cost $\kappa > 0$. If an optimal stationary policy switches from 0 to 1 only when $D(x) \geq \kappa$ and from 1 to 0 only when $D(x) \leq -\kappa$, then its entry threshold $x_+ = D^{-1}(\kappa)$ exceeds its exit threshold $x_- = D^{-1}(-\kappa)$.*

Proof. Strict monotonicity and $-\kappa < \kappa$ imply $D^{-1}(-\kappa) < D^{-1}(\kappa)$. $\square$

This proposition captures the elementary source of inertia. More general optimal-switching problems require assumptions about dynamics, observation noise, discounting, and regularity. It is not generally valid to claim that every optimal policy has a scalar hysteresis band or that band width must always increase monotonically with noise variance.

7.5 Safe blending

Theorem 3 (Convex safety of blended requests). *Let $S \subseteq C$ be a convex set of configurations satisfying all hard invariants. If each individually governed regime target c_e and the baseline c_0 lie in S , then every convex blend*

$$c = \beta_0 c_0 + \sum_e \beta_e c_e, \quad \beta_e \geq 0, \quad \sum_{e \geq 0} \beta_e = 1, \quad (15)$$

also lies in S .

Proof. This is the defining closure property of a convex set. $\square$

This gives a precise condition under which co-activation is safer than winner take all. It does not apply to nonconvex constraints such as discrete tool permissions, mutually exclusive modes, or minimum quorum requirements. Those coordinates require projection, arbitration, or exclusion by G .

7.6 Legibility bounds

Theorem 4 (Exact and expected legibility). *If an overseer must identify one active regime exactly from a worst-case audit message of at most B bits, then at most 2^B regimes can be distinguished. For a random active regime E encoded losslessly with expected length at most B by a prefix code, its entropy satisfies $H(E) \leq B$.*

Proof. There are at most 2^B distinct fixed-length bit strings. The expected-length claim follows from the source coding lower bound for prefix codes. $\square$

The result constrains distinguishable audit states, not necessarily the number of internal continuous states or regime types. Hierarchical queries, shared codes, approximation, and tolerated error change the bound. Legibility is a governance tradeoff, not a proof that five to seven emotions is universally optimal.

Conjecture 1 (Self-indexing under shift). *For task distributions in which utility depends on changing goals, capabilities, commitments, or authorization, an appraisal controller that uses calibrated OmegaSelf variables will transfer better under distribution shift than a controller restricted to otherwise matched world observations.*

This conjecture needs an explicit shift family. It is false in environments where world observations are sufficient statistics for optimal control, and it can fail when the self-model is stale or manipulated.

8 Candidate emotional constitutions

8.1 Research-assistant hive

The following names are mnemonic interface labels, not claims of human phenomenology.

Regime	Objectful appraisal	Candidate actuator request
Curiosity	A surprising, decision-relevant pattern with credible evidence and high expected information value	Increase query diversity and exploratory temperature; fund alternative hypotheses and bounded exploratory subagents.
Doubt	A high-impact claim with inconsistency, weak provenance, poor calibration, or missing evidence closure	Raise verification depth and source diversity; invoke checkers; lower confidence gain; defer irreversible output.
Frustration	A valued goal whose observed progress persistently falls below its forecast, with an attributed blocker or failing method	Raise plan repair, representation change, decomposition, or targeted escalation; limit repeated execution of the same method.
No-path distress	A valued goal with low progress and no currently conceived path, rather than a removable blocker	Seek assistance, expand the action set, revise deadlines or subgoals, or propose governed abandonment.
Consolidation	A result that is coherent, independently checked, and integrated with existing commitments	Shift resources to provenance, memory consolidation, ontology/Hyperseed writing, and reusable skill formation.
Boredom	Low marginal information gain and flat progress in a repetitive basin without urgent commitments	Reduce local allocation and expose background goals; never silently abandon a required task.
Integrity alarm	A mismatch among claim, evidence, action, authorization, and remembered commitments	Halt consequential action, preserve evidence, and invoke a policy or human review path.

“Frustration” should not merely increase generic arousal. Identified blocker, causal attribution, and alternative reachability determine whether the useful response is targeted repair, broad exploration, help-seeking, or goal revision.

8.2 Cybersecurity hive

Regime	Appraisal	Candidate actuator request
Vigilance	Credible anomaly probability times asset value and possible blast radius	Increase monitoring, audit frequency, provenance requirements, and conservative permission requests; preserve explicit decay and false-positive feedback.

Adversarial search	An authorized red-team target with attack surface and bounded rules of engagement	Increase hypothesis diversity and probe planning inside an immutable authorization envelope.
Caution	High uncertainty and irreversibility near a consequential action	Raise quorum and evidence thresholds; prefer reversible containment.
Calibration repair	A false positive, missed detection, or misattributed cause with ground-truth feedback	Update detector and trust calibration; reduce the self-excitation gain of vigilance without suppressing auditability.
Trust/suspicion	Evidence about a particular agent’s reliability, provenance, independence, and incentives	Adjust report-fusion weights and verification, not global social condemnation; prevent unbounded trust-collapse cascades.

Emotional policy must never enlarge the authorization envelope. Red-team “aggression” changes search allocation inside permission; it does not grant a tool, target, or capability.

9 MetaMo/OpenPsi and OmegaClaw implementation

9.1 Observed source seam

At pinned MetaMo commit 102da717b5354d6fd49d9226d92a32091438bc94, motivational state is represented as goals plus modulators. The OpenPsi configuration includes valence, arousal, approach, resolution, selection threshold, and securing, and stimulus dimensions such as novelty, conduciveness, risk, and effort. Its feeling layer maps five modulator dimensions to fuzzy intensities for **happy**, **sad**, **angry**, and **fear**, then selects a dominant category above a threshold. The appraisal and feeling tests reproduce the configured equations.

This is a continuous state plus classification layer. The inspected feeling code does not itself define persistent per-emotion latch state, distinct entry and exit thresholds, an object/provenance record, or a governed OmegaClaw actuator bundle. MetaMo’s broader dynamics already provide damping, bounded updates, stability checks, subsystem composition, and persistent state within a running decision loop. The proposal should reuse these facilities rather than build a parallel motivational engine.

The current OmegaClaw GoalChainer bridge constructs MetaMo motivational states for decision cycles, but its interface uses positional vectors and does not yet make the proposed emotional state durable across independent OmegaClaw turns. The first integration improvement is therefore a named schema and an explicit state handoff, not a large palette.

9.2 Three-layer implementation

1. **Continuous substrate:** compute Psi needs and modulators from canonical OmegaSelf records and current stimuli.
2. **Objectful appraisal and latch:** create a provenance-bearing appraisal, compute regime evidence, and update entry, exit, decay, and coupling.

3. **Governed actuation:** translate modulation and latch state into bounded requests, then apply deontic, capability, and safety policy before ECAN, tools, memory, or subagents change.

The cognitive cycle is:

telemetry $\rightarrow$ *OmegaSelf* $\rightarrow$ objectful appraisal $\rightarrow$ Psi update
 $\rightarrow$ regime update $\rightarrow$ bounded requests $\rightarrow$ policy governor
 $\rightarrow$ ECAN/tools/subagents $\rightarrow$ receipts and credit.

Emotion should generally modify ECAN stimulus wages or policy parameters rather than overwrite short-term importance directly. A schematic wage is

$$w(a) = w_0(a) [1 + z_{cur}N(a) + z_{doubt}Q(a) + z_{frus}G(a)], \quad (16)$$

where N, Q, G measure novelty, epistemic relevance, and involvement in the stalled goal. ECAN's budget, rent, forgetting, and spreading dynamics remain responsible for the final attention allocation.

9.3 MeTTa-shaped schema

The following is an interface sketch, not syntax claimed to run unchanged on the pinned MetaMo commit.

```
(EmotionRegime doubt
  (AppraisalClass epistemic-inconsistency)
  (Activation 0.0)
  (EnterThreshold 0.70)
  (ExitThreshold 0.35)
  (Decay 0.12)
  (ObjectSchema
    (goal claim evidence attribution controllability alternatives outcome))
  (Requests
    (verification-depth 1.8)
    (source-diversity 1.5)
    (irreversible-action-threshold 1.6)
    (confidence-gain 0.7)))

(RegimeReceipt doubt
  (Trigger appraisal-id)
  (EvidenceClosure evidence-set-id)
  (Requested request-id)
  (Applied governed-delta-id)
  (Outcome outcome-id))
```

The receipt records requested versus applied effects. This distinction is essential when a clamp or permission gate blocks part of a bundle.

9.4 Safety invariants

The governor should enforce at least:

1. emotional requests cannot expand tool permissions or target scope;

2. a regime cannot disable its own appraisal audit, provenance capture, or incident receipt;
3. consequential verification has a non-reducible floor;
4. all deltas, rates, and dwell times are bounded and logged;
5. discrete nonconvex conflicts are arbitrated, not naively averaged;
6. closed-loop gain, occupancy, and recovery are monitored;
7. counterfactual appraisals cannot become live evidence without a typed transition; and
8. human interruption and policy override remain outside emotional control.

An “anti-deception theorem” cannot be obtained merely by asserting that a verification coordinate never decreases. Integrity is multidimensional, and a regime might starve auditors of compute, bias their evidence, or alter routing. The appropriate artifact is an assurance case over named invariants and tested consumers.

10 Staged research program

10.1 Phase 0: shadow continuous baseline

Implement the pinned OpenPsi/MetaMo dynamics in a persistent OmegaClaw state adapter and log their proposed effects without actuation. Reproduce upstream fixtures and create a versioned conformance suite. Feed needs from OmegaSelf records for task progress, epistemic integrity, resource pressure, commitments, and novelty. This phase validates units, ranges, time constants, missing-data behavior, and cross-turn state continuity.

10.2 The decisive latch ablation

On matched tasks compare:

1. continuous Psi modulation only;
2. continuous Psi plus descriptive emotion labels;
3. continuous Psi plus objectful, hysteretic regimes in shadow mode;
4. continuous Psi plus governed regime actuation on safe coordinates.

The primary contrast is (1) versus (4); (2) detects benefits caused only by reporting or prompting; (3) measures classification, persistence, and predicted counterfactual effects before live mutation.

Metrics should include long-horizon goal utility, calibration, evidence quality, verification failures, time and tool cost, switching rate, stall-recovery time, unnecessary persistence, active-regime occupancy, explanation accuracy, and the overseer’s time or bits needed to identify the operative control state.

10.3 Targeted experiments

Object and attribution test. Hold negative valence and goal discrepancy approximately constant while varying whether a blocker is identified, attributed, controllable, and associated with a conceived path. Test whether distinct policies (targeted repair, exploration, help-seeking, or goal revision) improve outcomes compared with a valence-only controller.

Tempo-separation test. Present tasks with similar immediate reward histories but different persistent causal structure. Test whether the regime state captures useful slow context beyond continuous fast modulators. This is a direct test of temporal abstraction, not of emotion labels.

Hysteresis/noise test. Vary appraisal noise and switching cost. Estimate entry and exit behavior, chattering, delay, and utility. Compare learned boundaries against zero-hysteresis and fixed-band controls rather than assuming monotonicity.

Closed-loop gain test. Intervene on activation and measure how much a regime changes its own future appraisal input. Fit local Jacobians or impulse responses, validate held-out recovery predictions, and attenuate a bundle when estimated gain threatens the contraction margin.

Palette construction test. Start with a small set of operationally distinct regimes, initially curiosity and frustration or doubt and consolidation. Permit splitting only when contexts with the same label require reliably different bundles; permit merging when two regimes are behaviorally and predictively redundant. Palette size is selected by out-of-sample value plus description/maintenance cost, not folk-psychological completeness.

10.4 Rollout gates

Shadow	Compute and log needs, modulators, appraisals, regime states, counterfactual requests, and receipts. No behavior changes.
Advisory	Expose regime state as a soft utility/context feature to MetaMo ranking. Restrict effects to reversible, low-risk coordinates.
Governed actuation	Enable only bundles with measured behavioral support, positive held-out causal value, acceptable recovery, and complete audit records.
Adaptive palette	Learn weights or split/merge regimes only behind a versioned proposal, offline replay, safety review, and rollback path.

Soft bias is the safer default under appraisal noise. Hard preconditions are appropriate only for policy facts such as authorization, not for uncertain emotional categorization.

11 Failure modes and open questions

- **Decorative affect:** labels change prose but no behavior kernel. Proposition 1 predicts no instrumental gain.
- **Anthropomorphic mismatch:** human labels import animal action assumptions absent from the agent’s ecology.
- **Appraisal poisoning:** stale, correlated, adversarial, or imagined evidence activates a real bundle. OmegaSelf evidence closure and source diversity are central defenses.
- **Self-confirming regimes:** vigilance raises detector sensitivity, which increases anomalies, which raises vigilance. Monitor closed-loop gain, not decay in isolation.

- **Complacency:** satisfaction suppresses search before independent reproduction. Consolidation must have explicit evidence criteria.
- **Compulsive doubt:** verification consumes the entire budget. Apply bounded dwell, value-of-information stopping, and human escalation.
- **Trust collapse:** correlated reports are treated as independent evidence and suspicion spreads globally. Trust must be object-specific and dependence-aware.
- **Reward hacking and self-protection:** a regime changes its own success metric, hides receipts, or resists termination. Metrics and override mechanisms must sit outside the affected bundle.
- **Semantic drift:** positional MetaMo vectors silently change meaning when dimensions are added. Use named, versioned schemas.
- **Over-ontology:** latches add complexity without predictive or control value. The continuous-only ablation can falsify their necessity.

Open research questions include causal credit assignment over long dwell times; composition of social trust regimes across agents; learning objects and attributions; principled time-scale separation between mood and episode; robust self-appraisal under distribution shift; and whether an inspectable macrostate improves human governance enough to justify performance costs.

12 Conclusion

An OmegaSelf-compatible theory of emotion should begin from the agent’s ecology, not from a list of mammalian labels. Psi provides a serious continuous account of needs, value, competence, modulation, object direction, and action readiness. OmegaSelf makes the relevant self-appraisals evidence-bearing, contextual, predictive, and governable. A Hyperseed-style emotional regime is then an optional temporal macro-operator: it latches an objectful appraisal into a persistent, legible, bounded pattern of configuration requests.

The resulting design principle is stronger and more cautious than “give the agent emotions.” Give the system continuous motivation; represent what the appraisal is about; let every proposed regime name real behavioral levers; place all effects behind policy; measure recovery and self-excitation; preserve receipts; and retain a latch only when a controlled comparison shows that it improves long-horizon control or auditability. That is enough to make emotion a scientifically useful architectural hypothesis without claiming that OmegaClaw feels.

A Assumption checklist for the formal claims

Result	Required assumptions often omitted in informal statements
Actuator nullity	Equality of behavior kernels over the evaluation horizon; no direct utility/observation channel; positive cost for strict negative value.
Return to baseline	Finite palette; bounded effects; no re-evocation; continuous governor; continuous substrate returns to its baseline.

Contraction	A norm in which the closed-loop map is contractive; bounded input; nonexpansive clipping.
Hysteresis	Two regimes; scalar sufficient statistic; monotone continuation advantage; symmetric positive switching cost; threshold-form optimal policy.
Safe blending	Convex safety set and convex combination; excludes discrete or otherwise nonconvex constraints.
Legibility	Exact worst-case identification for the cardinality form; lossless prefix coding for the expected entropy form.

B Artifact and source provenance

- Bach tutorial: <http://bach.ai/pub/AAAI2018%20Modeling%20emotion%20and%20motivation%20tutorial.pdf>, retrieved 19 July 2026; locally preserved SHA-256 (line-wrapped): 90a40cb19ee824aa3ed46d8171de78cb35f89d6f2e7492977c739a74f475e225.
- MetaMo: <https://github.com/iCog-Labs-Dev/MetaMo.git>, commit 102da717b5354d6fd49d9226d92a32091438bc94; sixteen upstream MeTTa test files passed locally through the pinned OmegaClaw PeTTa environment.
- OmegaSelf Architecture and Deployment Guide, first public edition, 15 July 2026; locally preserved SHA-256 (line-wrapped): d8f6da10781f00e843dad3cf564405672bda522715b71aa242364ea242b9cc13.
- The original discussion-derived note was provisionally numbered 0014, but 0014 was already occupied by the OmegaSelf failure-taxonomy note. This systematic paper therefore uses paper number 0015.

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